

Health Research from Home (HRfH)

Hackathon 2026



Participant Manual and Guidelines

Introduction

Welcome to the Health Research from Home (HRfH) Hackathon 2026, hosted by the University of Manchester.

This hackathon aims to bring early career researchers together with an experienced faculty including public contributors to collaboratively explore innovative solutions and practical tasks in health research using smartphones and wearables.

The event emphasises building a collaborative community that enables participants to engage deeply with real-world health challenges through multidisciplinary teamwork, fostering connections, and creative problem-solving. This manual serves as your guide, providing essential information on navigating the event, understanding your tasks, accessing resources, and maximising collaborative outcomes. We look forward to seeing you all in Manchester!

Contact us: hrfh@manchester.ac.uk

Teams channel: [Teams](#)

Teamwork folder: [SharePoint Folder](#)

Project submission folder: [Submission](#)

Wi-Fi access: [Wi-Fi Info](#)

Event Schedule

The hackathon will be held in the Emmeline Suite, on the ground floor of the [Christabel Pankhurst Institute](#), University of Manchester (Dover Street, M13 9PS), from Wednesday 13th May to Friday 15th May, 2026.

Wednesday 13 th May	
Registration & breakfast	9:00 – 9:30
Introductory talks by Prof. Will Dixon: • Introduction to Health Research from Home • Introduction to hackathon team	9:30 – 9:45
Ice breaker & team formation	9:45 – 10:30
Task 1 background and Task 1 definition	10:30 – 10:45
Patient and Public Involvement and Engagement (PPIE) talk: • Lynn Laidlaw – Involving patients and the public in missing data; Purpose, provocations and reflexivity • Matt Barker, Colette Kelly and Lissie Fahey	10:45 – 11:15
Groupwork	11:15 – 12:45
Lunch (Provided)	12:45 – 13:30
Invited talk: Dr. Alexander Pate – Practical Design and Clear Reporting of Simulation Studies	13:30 – 14:15
Group work	14:15 – 16:45
End of day meeting, troubleshooting & planning for day 2	16:45 – 17:00
Evening meal (Provided) • Location: Bundobust at St James Building, 61-69 Oxford Street, Manchester, M1 6EQ	19:00 – 21:00
Thursday 14 th May	
Breakfast	9:00 – 9:30
Group work	9:30 – 12:30
Task 1 submission	12:30
Task 2 definition	12:30 – 12:45
Lunch (Provided)	12:45 – 13:30
Invited Talk:	13:30 – 14:15

Dr. Mia Tackney – Mind the Gap: Understanding, Defining and Handling Missing Accelerometer Data	
Group work	14:15 – 16:45
End of day meeting, troubleshooting & planning for day 3	16:45 – 17:00
Friday 15th May	
Breakfast	9:00 – 9:30
Group work	9:30 – 12:30
Task 2 and final slides submission	12:30
Lunch (Provided)	12:30 – 13:30
Final group presentations - Each team should present both Task 1 and Task 2 within a single presentation. - Presentation format: 15 minutes presentation + 5 minutes Q&A per team	13:30 – 15:20
Coffee break	15:20 – 16:00
Result & Group feedback	16:00 – 17:00

Task Descriptions

This hackathon focuses on recovery after treatment, such as knee replacement surgery in people with knee osteoarthritis. Knee osteoarthritis is a long-term degenerative condition that often causes pain and reduced mobility before surgery. As a result, daily activity levels, such as step count, are often already lower before the treatment. After surgery, recovery is gradual and varies across people. Some people recover quickly, while others recover more slowly or show more variable patterns over time. This makes the recovery process different from that of a person recovering from a short-term injury.

The aim of this hackathon is to explore both how to simulate realistic recovery patterns and how to analyse them when data are incomplete. The two tasks are linked but have different purposes. Task 1 focuses on building a realistic simulator of post-treatment recovery and missingness. Task 2 focuses on analysing the simulated data and exploring the impact of missingness on the findings.

Task 1: Simulation of post-treatment wearable/smartphone trajectories with realistic missingness

In Task 1, you will build a reusable simulator that generates synthetic activity recovery trajectories using daily step count after treatment, such as after knee replacement surgery. By reusable, we mean that your simulator should be clearly structured,

transparent, and easy to adapt to different assumptions, recovery patterns, levels of variability, or missingness scenarios. Your simulator should produce 1) realistic group recovery patterns, 2) realistic individual day-to-day variation, and 3) realistic missingness that resembles wearable or smartphone data collected in real life.

The aim is not to create a single “correct” recovery model. The aim is to design a simulator that is believable, clearly explained, and useful for later analysis.

Your simulator should generate trajectories of daily step counts for a year (365 days) post-treatment. Day 0 is the treatment date. A simple pre-treatment baseline period is recommended, especially if you want to use baseline information in later analysis. If you include a baseline period, keep it simple and explain how it is used.

Your challenge

Design a realistic simulation with the following core features.

Recovery patterns

Your simulator should generate trajectories using daily step count that reflect meaningful differences in recovery patterns after treatment.

At a minimum, include three recovery groups with clearly different overall patterns. For example, these may represent faster, moderate, and slower recovery, although you may define them differently if you justify your choice.

There is no single correct model for a recovery curve. You may use logistic curves, piecewise curves, spline-based patterns, or another sensible approach. Briefly justify your choices using clinical reasoning, references, and/or PPIE input.

Simulator Requirements

Each simulated patient should have:

- A plausible overall recovery shape over time (using daily step count),
- Realistic day-to-day variation,
- Realistic between-person differences within each group, so patients in the same group are still not identical.

Realistic missingness

Your simulator must generate realistic missingness patterns that resemble what is commonly seen in wearable or smartphone data.

Missingness should be a configurable part of the simulator, so that different scenarios can be generated and compared.

Teams should think carefully about:

- Why data may be missing,
- How much data is missing,

- How missingness is distributed over time,
- Whether missingness may relate to behaviour, recovery, or device use,
- Whether missingness appears as isolated days, short gaps, or longer continuous gaps,
- How missingness might vary between simulated individuals,
- What perspectives are necessary to facilitate our understanding of missingness.

The aim is not only to “remove data,” but to produce missingness patterns that are plausible, explainable, and reflect realistic wearable/smartphone data loss in practice, and reflect the lived experience of people using these devices.

Visualization

To verify and explain your simulator, provide figures or tables for a subset of simulated patients.

- Example trajectory plots (using step counts) covering each recovery group, with at least 3 individuals per group.
- A missingness visualisation showing observed and missing sections on example patient-level trajectories.

If you use a baseline period or key events, clearly annotate them.

Reproducibility and usability

Your simulator should be implemented in an open-source language (R and/or Python) and be runnable end-to-end from a clean environment.

You should include:

- Clear and well-structured code,
- A README.md file with:
 - How to run the simulator,
 - How to change key parameters,
 - Expected outputs and file locations.

A notebook is acceptable if it runs from top to bottom with no disordered cells or hidden dependencies.

Evaluation criteria

- Clear rationale for simulator design choices and assumptions.
- Recovery patterns are believable and clinically sensible, with assumptions clearly explained and justified where possible.
- Variability is plausible both within and between individuals.
- Missingness patterns realistically reflect wearable or smartphone use, with assumptions clearly explained and justified where possible.

- The simulator is reusable, reproducible, and produces clear, good-quality outputs that are suitable for downstream analysis and aligned with open science principles.
- Presentation of the simulator, outputs, and assumptions is clear and easy to follow.
- Learning from PPIE and clinical input is incorporated appropriately and meaningfully.

Expected output format

Submit:

- 1) Simulator code,
- 2) A README.md file,
 - a. How to run the simulator
 - b. Required environment setup
 - c. How to change key simulator parameters
 - d. Expected output files and folder locations
- 3) Simulator outputs, including generated data, figures and/or tables prepared for the final presentation
 - a. You can revise your presentation slides after submitting Task 1. However, you must not change any figures, tables, or results generated by your simulator after the Task 1 submission deadline.

Clearly label your files and folders with your team's name for easy identification.

Task 2: Ability to identify recovery trajectory groups under different patterns of missingness

In Task 2, you will work with an organiser-provided online simulator interface. **The access link will be released via [Teams](#) on May 14th.** This simulator generates daily step count recovery trajectories after treatment – similar to your Task 1.

The simulator is designed to let you explore different missingness scenarios and generate incomplete datasets for analysis. You will be able to interact with the simulator through the online interface, but you will not be given the underlying simulator code.

The aim of this task is to explore how well recovery trajectory groups can be identified from a population dataset when data are partially missing, and how performance changes under different missingness settings.

Your challenge

Develop an analysis approach that 1) uses daily step counts to identify group-based trajectories, then 2) develops a clear assessment of how recovery group identification changes under different missingness settings.

You are free to choose your own analysis approach.

What to explore

Your analysis should consider the following questions:

- How well can the recovery groups be identified when there is no missingness or very little missingness?
- How does performance change as the amount of missingness increases?
- Does performance differ when missingness occurs in different ways?
- What level or type of missingness appears to make recovery group identification unreliable?

You should focus on generating an analysis that is clear, interpretable, and useful for health research.

Output and Visualisation

Please provide clear outputs that show how your results change under different missingness settings. You are free to decide which summaries, plots, or comparisons are most useful for demonstrating your approach and findings.

Your figures should be clearly labelled and easy to interpret. If you use baseline information or any other factors in your analysis, clearly annotate them on the relevant figures.

Reproducibility and usability

Your Task 2 analysis should be implemented in an open-source language (R and/or Python) and be runnable end-to-end from a clean environment using the organiser-provided simulator outputs.

You should include:

- Clear and well-structured code,
- A README.md file with:
 - How to run the analysis,
 - What inputs are required,
 - How key comparisons are produced,
 - Expected outputs and file locations.

A notebook is acceptable if it runs top to bottom with no disordered cells or hidden dependencies.

Evaluation criteria

- Whether the approach used to identify recovery groups is appropriate for the question and the data,
- Whether the approach used to evaluate performance of the above method(s) under different missingness scenarios is appropriate and clearly justified,
- Whether the outputs clearly show how missingness affects recovery group identification, with clarity and appropriateness of interpretation,
- Identification of missing data scenarios where caution is needed, with practical recommendations for analysis or interpretation,
- Learning from PPIE and other relevant input is incorporated appropriately and meaningfully,
- Clarity and quality of the overall presentation.

Expected output format

Submit:

- 1) Analysis code,
- 2) A README.md file,
- 3) Final presentation slides

Clearly label your files and folders with your team's name for easy identification.

Submission Guidelines

The deadlines:

- **Task 1** (simulator) deadline: **12:30 on Day 2 (May 14th)**
- **Task 2** and **Final presentation slides** deadline: **12:30 on Day 3 (May 15th)**
 - Final Presentation: Each team should prepare slides to present the results from Task 1 and Task 2 in a structured way. An expert panel will evaluate and score the work based on the presentation, clarity, realism of the simulated data, and quality of the data analysis. The presentation should be no longer than 15 minutes, followed by a 5-minute Q&A session.
 - To help the panel follow your work, please briefly cover:
 - Simulator design and key assumptions
 - Simulated recovery patterns, variability, and missingness
 - Demonstrate the reusability of your simulator
 - Task 2 analysis approach

- Main findings on the effect of missingness
- Key limitations or areas where caution is needed.

Submit your work here: [Submission](#)

Usage Rights for Submitted Project

All code and other materials created during the Hackathon event by you shall be referred to as your Results. By submitting your Results, you grant The University of Manchester (“University”) an unrestricted worldwide, royalty-free, non-exclusive, irrevocable license to use, modify, distribute, and make your Results available on open-source platforms such as GitHub. This license includes the right for the University to use, adapt and share with third parties your Results for academic teaching and research purposes.

Contact and Support

- Event Leads: Dr. Shuai Shao & Prof. William Dixon
- Event Admin: Coral Stevenson & Louise Cook
- Expert Panel: Prof. William Dixon, Dr. Alex Pate and Dr. Mia Tackney
- Facilitators: Dr. Shuai Shao, Dr. Emma Pritchard, Matilda Pitt, Maureen Ngetich and Rebecca Gordon
- Public Contributors: Lynn Laidlaw, Matt Barker, Colette Kelly and Lissie Fahey
- Comms: Fu Lian Doble